

Geospatial Kernel: a state–action–constraint core for urban land-cover planning

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Geospatial Kernel: a state–action–constraint core for urban land-cover planning

An Abu Dhabi benchmark with conditional multi-year scenario analysis

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Abstract

Urban land-change models test how development demand may be distributed across space. Their planning value depends on explicit actions, hard constraints and consistent multi-year state transitions. We examine Geospatial Kernel, the algorithmic core of a Geospatial World Model, as an auditable state–action–constraint transition system for urban land-cover allocation. The public Abu Dhabi benchmark uses annual land-cover products from 2017–2024, a 100-m grid and three computational seeds, with GeoSOS-FLUS and GeoFM-LDN as comparison pipelines. Following peer review, the evaluator now prespecifies destination-correct multi-class change Figure of Merit, persistence and random-feasible controls, paired-pixel bootstrap intervals, and morphology objectives that are not direct scenario inputs or allocator terms. The archived binary-change run suggested different leaders at the one- and two-step horizons, but those rankings are not reissued as current evidence because the reviewed checkout lacks the complete aligned raster bundle, GeoFM-LDN checkpoint, FLUS runtime and independent change-validation data. The earlier planning Pareto claim is withdrawn pending recomputation. The implemented contribution is therefore an inspectable proposal–projection–writeback execution boundary and a fail-closed evaluation protocol. A defensible resubmission requires a matched-input FLUS experiment, independent validation of 2023–2024 change, and regeneration of all numerical tables and figures. The current evidence does not establish a universal accuracy ranking or an Abu Dhabi forecast.

Highlights

- Geospatial Kernel separates transition proposals from constraint projection.
- Archived model rankings await strict-metric and matched-input reruns.
- The earlier planning Pareto claim is withdrawn pending recomputation.
- Public-data maps are stress tests, not statutory Abu Dhabi forecasts.

Keywords

Geospatial World Model; land-cover change; constrained allocation; cellular automata; Abu Dhabi; scenario planning

Introduction

Land-cover change is a spatial process. Development demand is expressed as a quantity, but its consequences depend on where each new cell is placed relative to roads, existing built areas, ecological features and other land-cover classes. This coupling matters for urban planning because two maps with the same total built area can imply very different infrastructure requirements and environmental pressures. Cellular-automata and suitability-based models have long provided a way to distribute land demand across a map, including CLUE-S, SLEUTH, Dinamica EGO, FLUS, local-attribute urban CA models and more recent patch-generating or deep-learning CA models (Verburg et al., 2002; Silva and Clarke, 2002; Soares-Filho et al., 2002; Lau and Kam, 2005; Liu et al., 2017; Liang et al., 2021). Global Earth-observation products now make it possible to construct annual, spatially aligned inputs for cities that lack a harmonized local data archive (Brown et al., 2022; Gorelick et al., 2017).

Validation studies have also shown that agreement in unchanged cells can mask poor allocation of actual change. Quantity and allocation disagreement, destination-specific transition scores and explicit calibration/validation protocols are therefore preferable to a single overall-accuracy number (Pontius et al., 2008; Pontius and Millones, 2011; van Vliet et al., 2011, 2016). This distinction is central here because the benchmark uses noisy annual labels and tests both one-step allocation and recursive rollout.

The main technical difficulty is not producing another categorical raster. It is keeping the semantics of a planning action separate from the learned transition tendency. A model may predict that a cell is likely to become built, yet the resulting map can still violate a water exclusion, miss the requested class totals or become inconsistent when the next year is simulated from the model’s own output. These failure modes are particularly important when the model is used to compare scenarios rather than to reproduce one observed transition. Latent-dynamics approaches such as GeoFM-LDN can represent continuous geospatial state and condition transitions on demand, whereas traditional ANN-plus-cellular-automata systems such as GeoSOS-FLUS combine suitability with explicit neighbourhood evolution. Neither design, by itself, guarantees that action semantics, hard constraints, state persistence and evidence provenance are visible at the same execution boundary.

We address this systems problem by studying Geospatial Kernel, the core transition layer of a Geospatial World Model (GWM). The Kernel represents one simulation step as a typed state, a typed action, a transition proposal, a constraint projection and a state writeback. The learned proposal remains domain-specific; the runtime contract is domain-neutral. For Abu Dhabi land-cover planning, the proposal is a six-class probability cube, and the projection allocates only mutable cells until the action’s feasible class totals are reached. This separation makes the planning operation inspectable: a reviewer can distinguish what the transition model preferred from what the constraints admitted.

We evaluate this design in a controlled public-data benchmark rather than treating the benchmark as an official data release. All candidates consume the same 100-m grid, common valid mask, annual

class-count actions, hard exclusions and evaluator. The historical track uses observed target counts to isolate spatial allocation skill from demand error. The planning track starts from the observed 2024 state and applies three planner-supplied actions: moderate growth, green-priority growth and high outward growth. Abu Dhabi Plan 2030 provides an institutional planning context for discussing urban structure, but the numerical actions in this benchmark are not calibrated to that plan or to an official population forecast (Abu Dhabi Urban Planning Council, 2007). We ask four questions: whether the Kernel execution contract is auditable, whether its allocation skill is competitive, whether it produces feasible and spatially distinct allocations and which conclusions survive label and parameter sensitivity.

The paper makes three bounded contributions. First, it gives a concrete state–action–constraint formulation for using a geospatial kernel as an algorithmic core rather than as an opaque end-to-end predictor. Second, it reports a multi-horizon pipeline comparison with GeoSOS-FLUS and GeoFM-LDN, including a case in which the Kernel does not win, while making the input imbalance explicit. Third, it turns the future maps into explicit stress-test artefacts, including cumulative rasters and vectorized transition footprints, while preserving the distinction between public-data land cover and authoritative legal land use.

Methods

Study boundary, grid and temporal protocol

The study boundary is the polygon returned for OpenStreetMap relation R4479763, representing Abu Dhabi city. All rasters were aligned to a canonical 100-m grid in EPSG:32640 with 475 columns and 360 rows. A cell was included when its centre fell within the city polygon and when the common public-data mask indicated valid coverage. The benchmark profile records the source geometry, raster transform, dimensions and SHA-256 hashes. The effective evaluation mask contains 79,726 cells.

The observed sequence contains annual states for 2017–2024. Models were fit using the four transitions 2017→2018, 2018→2019, 2019→2020 and 2020→2021. A 2021→2022 transition was used for validation during model development. The historical test starts from 2022 and evaluates 2023 and 2024 in an open-loop rollout without observed-state writeback. The planning test starts from the observed 2024 state and recursively generates 2025–2031 under each scenario action.

Public data and class semantics

Dynamic World V1 provides annual scene-argmax labels and mean top-class probabilities (Brown et al., 2022). The nine Dynamic World labels were crosswalked to six canonical classes: water (source 0), woody vegetation (1), low vegetation (2, 4 and 5), wetland (3), built (6) and bare (7). Source class 8 was excluded. The resulting label is explicitly treated as land cover. No residential, commercial or industrial category is inferred from the raster labels.

Annual 64-dimensional AlphaEarth embeddings were spatially averaged to the canonical grid and locally L2-normalized. Monthly VIIRS night-time-light radiance was averaged by year. Copernicus DEM supplied elevation and a finite-difference slope derivative. OpenStreetMap road geometries were rasterized into distance-to-road and distance-to-major-road surfaces. ESA WorldCover 2021 supplied public water and wetland/mangrove constraint proxies (Zanaga et al., 2022). All dynamic and static layers were checked for exact CRS, transform, width and height agreement.

Geospatial Kernel state, action and transition proposal

Let $S_t(i)$ denote the six-class state at valid cell i , A_t the action for the interval $t \rightarrow t + 1$, and X_t the aligned driver layers. The Kernel separates a proposal from a projection:

$$Q_t(i, c) = p_\theta(c \mid x_i(S_t, X_t)), \quad S_{t+1} = \Pi_{C, A_t}(Q_t, S_t).$$

The feature vector x_i contains 25 values: six one-hot current-class indicators; twelve neighbourhood proportions, comprising 3×3 and 7×7 windows for each of the six classes; and seven continuous variables, namely normalized row and column coordinates, clipped elevation, clipped slope, log-transformed VIIRS radiance, log distance to roads and log distance to major roads. A histogram gradient-boosting classifier estimates the six-class probability vector. It uses learning rate 0.08, 120 iterations, at most 31 leaf nodes, minimum leaf size 40 and L2 regularization 1.0.

Training samples combine all changed valid cells and a 20% random sample of stable valid cells in each fit transition. The sample weight is the product of a change emphasis, $1 + 5\mathbf{1}(S_t \neq S_{t+1})$, and a confidence factor, $0.5 + \min(r_t, r_{t+1})$, where r is the Dynamic World mean top probability. The target years 2023 and 2024 are not used during fitting or model selection. Across the three seeds, each fitted model used 71,892–72,202 pixel rows and all six classes.

Constraint projection and state writeback

For a mutable source cell i of class s and target class c , the base allocation score is

$$q_{i,s \rightarrow c} = \log p_\theta(c \mid x_i) - \log p_\theta(s \mid x_i) + \lambda \rho_{i,c}^{(7)},$$

where $\rho_{i,c}^{(7)}$ is the 7×7 fraction of neighbouring cells currently in class c , and $\lambda = 0.35$. The algorithm computes class deficits and excesses from the action’s feasible target counts, ranks all admissible source–target candidates by q , and changes cells in descending order until all deficits and excesses are zero. Permanent water and the protected wetland/ecological mask are held fixed. If the counts cannot be satisfied without changing a hard-exclusion cell, the step fails closed rather than silently violating the action.

The projected raster is written as the next **KernelState**. Every step records a state reference, action evidence, model version, parameter reference, projection status and diagnostic counts. The runtime checks domain identity, source-time consistency and action time advancement. This execution contract is the Geospatial Kernel’s algorithmic contribution; the Abu Dhabi adapter supplies the land-cover-specific features, learner and constraint masks.

Baselines and common evaluator

GeoSOS-FLUS was run through the external FLUS console using seven continuous drivers: normalized row and column coordinates, elevation, slope, VIIRS radiance, distance to roads and distance to major roads. Its ANN suitability surface was passed to a cellular-automata allocation stage with the common class totals and public hard mask. The archived run used one 2021 training year, eight ANN hidden neurons, a 3×3 CA neighbourhood, neighbourhood strength 1.0, acceleration factor 0.1 and 1,000 iterations. Its frozen transition matrix protected water and wetland but did not prohibit built-to-non-built transitions; this configuration is recorded as a baseline limitation rather than a calibrated Abu Dhabi model.

GeoFM-LDN (Geospatial Foundation-Model Latent Dynamics Network) was trained as a demand-conditioned latent-dynamics network on AlphaEarth embeddings. A logistic semantic decoder maps the predicted embedding to six class probabilities, and the constrained allocation step converts those probabilities to a categorical raster. The latent and categorical states are both recursively carried into the next year. GeoFM-LDN is an internal benchmark implementation in this manuscript; the name describes its architecture and does not attribute an external publication or performance claim.

The common evaluator computes actual class counts, normalized demand total variation, hard-mask violations, strict multi-class change FoM, binary change FoM, change F1, overall accuracy and macro-F1 for historical runs. The strict FoM counts a hit only when the predicted destination class equals the observed destination class; wrong destination changes are retained in the denominator. It also generates persistence and random-feasible zero controls and paired-pixel bootstrap intervals for strict FoM, overall accuracy and macro-F1. For planning runs it additionally computes ecological conversion, new-built neighbourhood fraction, built component counts, built components per 1,000 valid pixels, a 500-m leapfrog rate, distances to roads and prior built cells, infrastructure proxy distance and built retirement. Pareto membership is calculated only over the four independent morphology/accessibility diagnostics: major-road distance, prior-built distance, built component density and leapfrog rate.

Scenario actions and uncertainty

The moderate-growth, green-priority-growth and high-outward-growth actions are stored in the tracked `planning_scenarios_public_2025_2031.json` manifest; the legacy `compact` path identifier is retained only for artifact compatibility. Each action supplies feasible class totals for 2025–2031. The 2031 totals are an explicit extension of the previous annual difference and are not derived from an official population, housing or infrastructure forecast. Future exogenous rasters are held at their 2024 values. The green-priority action is not a water-budget model. Each candidate is run at seeds 31, 47 and 73, and planning tables report arithmetic means and population standard deviations. No p-values are reported because the three seeds describe computational variability, not independent field samples.

Change polygons and reproducibility

For each model–scenario pair, annual and cumulative differences from the 2024 raster are extracted as changed 100-m cells and dissolved into GeoPackage layers. The public delivery manifest covers 45 ensemble rasters for 2027–2031 and nine vector packages. The audit script now reports `INCOMPLETE_INPUTS` or `FAIL` when source rasters, reports or vectors are absent; a `PASS` cannot be inferred from a stale audit file. The vector layers preserve raster-cell geometry and should not be interpreted as cadastral parcels.

Results

A common benchmark separates allocation skill from demand assumptions

The benchmark covers the Abu Dhabi city polygon represented by OpenStreetMap relation R4479763. The canonical grid is a 475×360 lattice in EPSG:32640 at 100-m resolution. After the common coverage and alignment checks, 79,726 cells were used for scoring. Each cell represents 1 ha. Annual Dynamic World observations from 2017 to 2024 were harmonized to six land-cover classes: water, woody vegetation, low vegetation, wetland, built and bare. AlphaEarth annual embeddings, VIIRS night-time lights, Copernicus DEM derivatives and OpenStreetMap road distances supplied transition drivers. ESA WorldCover and public OpenStreetMap geometries supplied water, wetland and infrastructure exclusion proxies.

The comparison shared the canonical grid, origin states, target actions, hard masks and evaluator, but the original public-data run was not information-balanced. GeoSOS-FLUS used seven continuous drivers in its external ANN suitability model, whereas Geospatial Kernel used 25 features including current-class indicators and neighbourhood proportions; GeoFM-LDN used AlphaEarth latent embeddings. We therefore treat the historical ranking as a pipeline comparison, not evidence that one learner is intrinsically superior. A matched-input FLUS rerun is required before making model-level superiority claims. All models used seeds 31, 47 and 73. The earlier output audit reported 240 seed and ensemble rasters for a separate 2025–2030 run; the public 2025–2031 delivery is a separate artifact set and must be audited separately.

The execution trace makes the distinction between proposal and admission explicit. A Kernel step records the source state, action, probability-cube proposal, projected state, model identity, input evidence and next-state reference. The runtime checks that the action advances the source time and that the projected raster becomes the next state. This is the operational meaning of the Kernel in this study: not a claim that all GWM domains share the same learner, but a claim that they can share an auditable execution contract. The shared inputs and execution boundary are summarized in Fig. 1.

Archived historical snapshot is horizon-dependent; current ranking is withheld

The historical track used observed target class totals. This choice removes one source of uncertainty, demand estimation, and tests how each model allocates the requested changes in space. The prespecified primary metric is a strict multi-class change FoM: a hit requires the predicted destination class to equal the observed destination class, while wrong destination changes remain in the denominator (Pontius et al., 2008). Binary change FoM, change F1, overall accuracy and macro-F1 are secondary diagnostics. Values are means with population standard deviations over the three seeds; paired-pixel bootstrap intervals are reported by the revised compiler when the raster artifacts are available.

The feasible target counts are not always identical to the observed target counts. In the archived action file, the fixed-water and protected-wetland projection changed the 2023 water target from 15,985 observed cells to 15,438 feasible cells and the 2024 water target from 16,060 to 15,438; the 2023 and 2024 built targets changed from 12,121 to 12,260 and from 15,598 to 15,774, respectively. Consequently, the historical task excludes reclamation-related water changes and its FoM cannot be interpreted as a full coastal-change score.

Table 1 | Archived historical conditional allocation snapshot (not inferential). The 2023 target is a one-step prediction from 2022. The 2024 target is reached by a two-step open-loop rollout from 2022 without observed-state writeback. These values are the earlier binary-change FoM artifact and are retained only for audit continuity; they must not be interpreted as the revised strict multi-class result. Persistence and random-feasible zero models, strict FoM and paired-pixel bootstrap intervals will replace this snapshot after the raster bundle is restored. DTV is normalized demand total variation.

Year	Method	Legacy binary FoM	Change F1	Overall acc.	Macro-F1	DTV
2023	GeoSOS-FLUS	0.1283 $\pm$ 0.0043	0.2273 $\pm$ 0.0068	0.9183 $\pm$ 0.0008	0.8441 $\pm$ 0.0034	0.002835
2023	Geospatial Kernel	0.2004 $\pm$ 0.0018	0.3338 $\pm$ 0.0026	0.9362 $\pm$ 0.0002	0.8499 $\pm$ 0.0003	0.000000
2023	GeoFM-LDN	0.1670 $\pm$ 0.0039	0.2862 $\pm$ 0.0058	0.9316 $\pm$ 0.0005	0.8457 $\pm$ 0.0006	0.000000
2024	GeoSOS-FLUS	0.1815 $\pm$ 0.0125	0.3071 $\pm$ 0.0180	0.8595 $\pm$ 0.0030	0.7483 $\pm$ 0.0062	0.005381
2024	Geospatial Kernel	0.2606 $\pm$ 0.0006	0.4134 $\pm$ 0.0008	0.8863 $\pm$ 0.0002	0.7670 $\pm$ 0.0004	0.000000
2024	GeoFM-LDN	0.2778 $\pm$ 0.0061	0.4348 $\pm$ 0.0074	0.8905 $\pm$ 0.0014	0.7693 $\pm$ 0.0011	0.000000

For transparency, the archived persistence control had overall accuracy 0.9436 and macro-F1 0.8684 for 2023, and 0.8938 and 0.7907 for 2024, respectively. These static values exceeded the learned candidates in the legacy artifact. They are not combined with the legacy model rows above because the strict-metric and bootstrap rerun is still blocked.

The archived binary-FoM snapshot showed Geospatial Kernel above GeoSOS-FLUS at the 2023 one-step horizon and GeoFM-LDN above both at the 2024 two-step horizon. Because those values use the old binary definition and omit the random-feasible control, they are descriptive continuity checks rather than evidence of a final ranking. The revised strict metrics, zero-model comparison and paired-pixel intervals are intentionally withheld until the source rasters are restored. The four primary metrics and seed variability are shown in Fig. 2 only as a legacy figure pending regeneration.

The reliability subset tells a second, less favourable story about the data. Dynamic World mean top-probability was below 0.5 for approximately 51.0% of valid cells in 2017 and 61.3% in 2024. When both origin and target cells were required to exceed 0.5, all models retained high static agreement but the change FoM became much lower because few high-confidence cells were labelled as changed. We therefore report the full-grid metric as the primary result and use the reliability subset only as a diagnostic of label volatility.

Archived planning snapshot illustrates scenario dependence (not inferential)

The planning track starts from the observed 2024 state and rolls forward to 2031. It uses three declared actions: moderate growth (the legacy **compact** path) adds 500 built and 50 low-vegetation cells per year; green-priority growth adds 350 built and 150 low-vegetation cells per year; and high outward growth adds 1,000 built and 25 low-vegetation cells per year. The action file contains class-count changes only; it does not encode a compactness objective, and the green-priority action does not represent a water-budget analysis. The 2031 targets extend the previous annual differences and are explicitly synthetic scenario targets. Exogenous drivers are held at their 2024 values, so the exercise is a controlled stress test rather than a forecast.

Table 2a | Archived 2031 planning snapshot (not inferential). Values are mean $\pm$ population SD across three seeds. DTV is normalized demand total variation; ecological conversion is the percentage of new built cells originating in the public vegetation/ecological proxy mask; neighbourhood fraction is descriptive only and is not a Pareto objective. Distances are measured in metres. The displayed compactness column is the legacy neighbourhood fraction and is not an independent morphology measure. The revised evaluator additionally reports built components per 1,000 valid pixels and a 500-m leapfrog rate; those values will replace this snapshot after rerun. **Kernel** denotes Geospatial Kernel.

Method	Scenario	DTV	Eco conv. (%)	Legacy neighbourhood fraction	Major road (m)	Prior built (m)
GeoSOS-FLUS	Moderate growth	0.00068	1.68	0.7618 ± 0.0231	446.1 ± 39.5	231.6 ± 21.3
GeoSOS-FLUS	Green-priority growth	0.00528	1.77	0.7687 ± 0.0291	443.7 ± 35.9	212.7 ± 11.4
GeoSOS-FLUS	High outward growth	0.00045	1.31	0.8000 ± 0.0196	443.5 ± 37.2	255.3 ± 27.8
Kernel	Moderate growth	0.00000	0.00	0.8410 ± 0.0045	334.7 ± 13.4	116.7 ± 1.2
Kernel	Green-priority growth	0.00000	0.00	0.8332 ± 0.0068	323.1 ± 19.5	111.0 ± 0.5
Kernel	High outward growth	0.00000	0.00	0.8545 ± 0.0047	365.5 ± 5.3	139.8 ± 1.5
GeoFM-LDN	Moderate growth	0.00000	0.00	0.7676 ± 0.0170	475.9 ± 35.7	239.7 ± 44.6

Method	Scenario	DTV	Eco conv. (%)	Legacy neighbourhood fraction	Major road (m)	Prior built (m)
GeoFM-LDN	Green-priority growth	0.00000	0.00	0.7506 ± 0.0174	452.1 ± 22.0	222.3 ± 19.8
GeoFM-LDN	High outward growth	0.00000	0.00	0.8117 ± 0.0236	507.5 ± 56.4	263.0 ± 71.8

Table 2b | Archived 2031 demand and transition snapshot (Pareto status withdrawn). Built gain is net built-cell change relative to 2024; retired counts cells that were built in 2024 and are not built in 2031; green gain is the corresponding net gain. These values are retained only for continuity. The earlier Pareto status used structural zero and scenario-input terms and is not evidence. The revised compiler uses only independent morphology and accessibility diagnostics and must be rerun from the raster artifacts.

Method	Scenario	Built gain	Retired	Green gain	Pareto status
GeoSOS-FLUS	Moderate growth	3,500	3,436	296	withdrawn
GeoSOS-FLUS	Green-priority growth	2,450	3,632	629	withdrawn
GeoSOS-FLUS	High outward growth	7,000	2,724	139	withdrawn
Kernel	Moderate growth	3,500	0	350	withdrawn
Kernel	Green-priority growth	2,450	0	1,050	withdrawn
Kernel	High outward growth	7,000	0	175	withdrawn
GeoFM-LDN	Moderate growth	3,500	0	350	withdrawn
GeoFM-LDN	Green-priority growth	2,450	0	1,050	withdrawn
GeoFM-LDN	High outward growth	7,000	0	175	withdrawn

The earlier statement that all three Geospatial Kernel candidates occupied a Pareto frontier is withdrawn for the revised manuscript because the original objective set included structural zeros, scenario inputs and a neighbourhood quantity used by the allocator. The revised objective set is implemented in `planning.py` and includes only mean distance to major roads, mean distance to prior built cells, built components per 1,000 valid cells and the independently defined 500-m leapfrog rate. The corresponding 2031 frontier must be regenerated from the public 2025–2031 raster artifacts. Ecological conversion remains a descriptive public-data proxy, not an estimate of statutory ecological impact.

The archived built-retirement contrast is visible in the legacy maps: GeoSOS-FLUS reclassified approximately 2,724–3,632 cells that were built in 2024 while allocating new built cells elsewhere, whereas Geospatial Kernel and GeoFM-LDN retained the prior built class under the same scenario target counts. This difference is a model behaviour under the frozen transition matrix, not evidence that either approach reproduces actual redevelopment or demolition. The revised objective trade-offs will be summarized in Fig. 3 after rerun.

Five-year trajectory and vector-footprint lineage (not inferential)

The Geospatial Kernel trajectories from 2027 to 2031 show how the declared class-count actions change the spatial configuration as demand accumulates. The earlier neighbourhood-fraction and distance

values are retained here as a legacy public-data snapshot, but they are not interpreted as proof that the moderate-growth action optimized compactness. The revised planning compiler adds built-component density and a 500-m leapfrog rate before any new frontier claim is made. Mean distance to pre-existing built cells is reported as a morphology diagnostic, not as a universally preferred planning value.

Table 3 | Archived Geospatial Kernel trajectory snapshot (not inferential). Values are three-seed means. Built and green gains are cumulative net changes relative to 2024. The scenario labels are descriptive; **Moderate growth** is the legacy compact path, and **Green-priority growth** does not include a water-budget constraint. Legacy neighbourhood fraction is shown for continuity only and is not a Pareto objective.

Scenario	Year	Built gain	Green gain	Legacy neighbourhood fraction	Major road (m)	Prior built (m)
Moderate growth	2027	1,500	150	0.7913	320.5	102.9
Moderate growth	2028	2,000	200	0.8067	324.5	105.6
Moderate growth	2029	2,500	250	0.8217	327.0	108.8
Moderate growth	2030	3,000	300	0.8318	331.7	112.5
Moderate growth	2031	3,500	350	0.8410	334.7	116.7
Green-priority growth	2027	1,050	450	0.7757	313.3	102.0
Green-priority growth	2028	1,400	600	0.8003	314.3	103.8
Green-priority growth	2029	1,750	750	0.8141	316.2	106.3
Green-priority growth	2030	2,100	900	0.8229	319.5	108.5
Green-priority growth	2031	2,450	1,050	0.8332	323.1	111.0
Outward growth	2027	3,000	75	0.7948	341.5	107.7
Outward growth	2028	4,000	100	0.8147	348.2	114.9
Outward growth	2029	5,000	125	0.8304	354.4	123.0
Outward growth	2030	6,000	150	0.8444	359.5	131.5
Outward growth	2031	7,000	175	0.8545	365.5	139.8

For delivery, each model and scenario has an ensemble raster for each year 2027–2031 and a vector package containing annual and cumulative transition layers. The nine GeoPackage packages contain polygons formed by dissolving changed 100-m cells; they are therefore raster-cell transition footprints rather than parcel boundaries. This distinction matters for downstream planning because the polygons can support visual review and overlay analysis, but they do not represent cadastral lots or legally approved projects. The 2031 state and transition footprints are visualized in Fig. 5.

Sensitivity shows a stable but not fully explained neighbourhood contribution

The base Kernel allocation score adds a 7×7 target-class neighbourhood fraction with weight 0.35. A post-hoc three-seed sensitivity run varied this weight from 0 to 0.7. The legacy binary-change FoM changed by less than 0.001 across the tested weights in both years. Because the neighbourhood fraction appears in both the proposal features and the allocation score, this sensitivity is not an independent test of compactness; the revised planning evaluator therefore excludes neighbourhood fraction from the Pareto objective set.

Mechanism controls separate learned proposal from runtime projection

We next used controlled variants to test which parts of the execution contract are consistent with the observed differences. These are mechanism controls, not causal policy experiments. The values retained from the original artifact are legacy binary-change FoM values and must be regenerated with the strict multi-class evaluator. Removing neighbourhood features from the proposal, removing continuous spatial drivers, shuffling those drivers, using a Markov proposal and using a random proposal all changed historical allocation skill, but they do not establish that any one feature causes real urban change.

Runtime controls changed the interpretation of the result. Deleting the action semantics reduced change FoM by 0.1161 in 2023 and 0.1760 in 2024. Shuffling the action had a small positive effect at the one-step horizon (+0.0071) but a negative effect in the two-step rollout (−0.0937), demonstrating that action sensitivity is horizon-dependent rather than monotonically harmful. Removing the allocation neighbourhood term changed FoM by less than 0.001 in either year, whereas removing state writeback had little effect in 2023 but increased the 2024 FoM by 0.0110; these controls therefore do not isolate a single source of historical skill.

The planning controls clarify why morphology cannot be attributed to the learned proposal alone. The Markov proposal paired with the same allocator was more compact than the full learner in the original artifact, showing that the shared projection and recursive state handling can dominate visual morphology. Deleting hard constraints introduced violations despite sometimes improving unconstrained proxy values. The revised interpretation is therefore limited: the controls support auditability and expose interactions among proposal, projection and writeback, but they do not identify a unique causal mechanism for a planning advantage. The full set of controls is shown in Fig. 4 and should be rerun after the strict evaluator and independent morphology metrics are materialized.

Discussion

This study positions Geospatial Kernel as a candidate execution layer for constrained spatial reasoning. The defensible result is not that one classifier wins every temporal test, but that a transition proposal and a planning admission rule can be represented and inspected separately. The archived per-seed public-data run reported feasible class-count allocations and zero declared hard-mask violations for the Kernel; the current checkout does not contain the aligned rasters needed to independently re-audit those outputs or the integrated delivery products. Claims about superior compactness or accessibility are withdrawn until the revised independent morphology objectives are rerun and the underlying raster artifacts are available.

The historical results place the planning experiment in context. The original binary-change FoM ranking favoured Geospatial Kernel at the one-step 2023 horizon and GeoFM-LDN at the two-step 2024 horizon, while the persistence zero model had the highest static accuracy. These values must be regenerated with the strict multi-class FoM and paired-pixel bootstrap before quantitative superiority claims are repeated. The benchmark does not support selecting one model as the universally most accurate future predictor.

The spatial objective differences are consistent with the design of the Kernel projection. The score compares the learned probability of a target class with that of the current class and adds local support from the target-class neighbourhood. The projection then resolves deficits and excesses while refusing mutable changes inside the hard mask. This makes it natural for the Kernel to retain existing built cells and to fill target classes near already compatible neighbourhoods. The allocator-matched controls above qualify this interpretation: planning morphology also depends on the shared projection and recursive state handling, and the present experiments do not identify a single causal pathway.

Several limitations define the boundary of the present evidence. The six classes are remote-sensing land cover, not statutory residential, commercial or industrial land use. Dynamic World labels show substantial annual uncertainty and apparent turnover, and the public road, wetland and protected-area layers are proxies rather than Abu Dhabi planning records. Future drivers are frozen at 2024, no observed 2025–2031 labels exist for validation and the scenario targets are synthetic planner-supplied actions. The three-seed summaries quantify stochastic variation but are not population-level uncertainty intervals. The completed controls are public-data stress tests with matched seeds rather than randomized interventions on real planning decisions, and several variants alter more than one implementation detail at a time. We therefore report them as evidence consistent with the execution contract rather than as causal attribution.

These limitations also indicate a practical path to deployment. An authoritative version would replace the Dynamic World class raster with a locally validated land-use/land-cover crosswalk, replace public proxy masks with statutory water, ecological, airport, port and growth-boundary layers, and supply approved development demand and infrastructure capacities. The Kernel contract can remain unchanged while those domain inputs are upgraded. This separation is useful for governance: a new data source changes the evidence attached to a transition, rather than silently changing the meaning of the execution trace.

Data availability

The public-data benchmark manifests, protocols, reports and generated delivery artefacts are organized under `benchmarks/abu_dhabi_land_use_v1/` in the accompanying repository. The main evidence files are:

- `comparison_report.json` and `comparison_report.md` for historical metrics;
- `planning_comparison_report_public_2025_2031.json` and its Markdown rendering for 2031 planning objectives;
- `planning_scenario_report_public_2025_2031.json` for per-seed rollout traces;
- `planning_public_2025_2031_delivery_manifest.json` for raster and vector delivery;
- `output_audit.json` for the versioned raster audit; the revised audit does not treat the earlier 240-raster 2025–2030 record as evidence for the separate 2025–2031 delivery;
- `data_audit.json`, `protocol.json`, `gee_input_manifest.json` and `osm_input_manifest.json` for data provenance;
- Large raster files are tracked through repository-relative paths and SHA-256 manifests rather than embedded in this manuscript. At the time of revision, the checkout used for review does not contain the complete `artifacts/gee`, `artifacts/bundle` or historical prediction rasters, so the experiment cannot be rerun from this repository alone. No private database address, credential or client-only service is required for the intended public-data benchmark, but the missing artifacts must be restored before a reproducibility claim is made.

Code availability

The analysis scripts, a vendored Geospatial Kernel runtime snapshot, benchmark protocol and figure-generation code are maintained in the accompanying repository. Principal entry points are listed separately to keep the manuscript readable:

- `run_geospatial_kernel.py` for the Kernel historical run;
- `data_agent/uwm/geospatial_kernel/runtime.py` for the vendored runtime snapshot;
- `render_nature_figures.py` for publication artwork.

The repository is currently private and has no archival DOI; a public URL, release tag and DOI must be supplied before submission. GeoFM-LDN code and checkpoints are not included in this repository and remain a reproducibility blocker.

Declarations

Funding. No external funding was received for this study.

Competing interests. Ning Zhou is employed by Beijing Freedo Technology Co., Ltd., which develops geospatial-world-model software. The author declares no other competing financial or personal interests.

CRedit author statement. Ning Zhou: Conceptualization, methodology, software, data curation, formal analysis, visualization, writing—original draft, writing—review and editing.

Acknowledgements. None.

Ethics statement. Not applicable; the study used geospatial raster, vector and derived public-data products and did not involve human or animal participants.

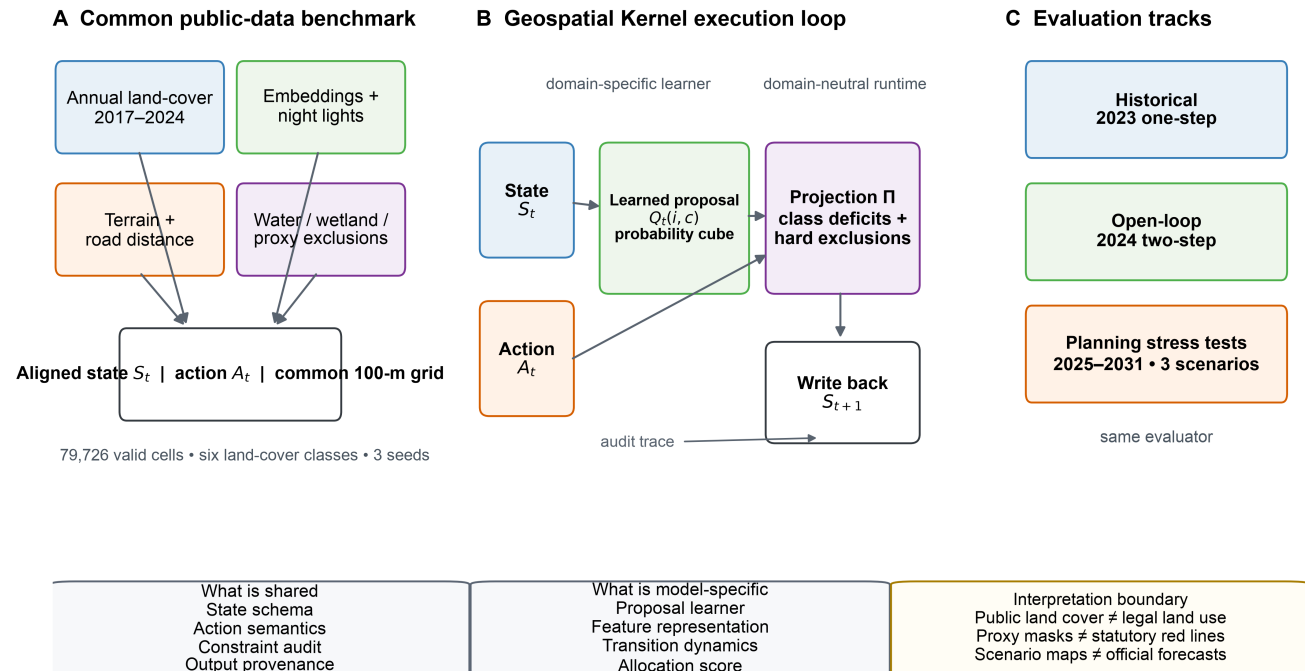
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Figure legends

A common state–action–constraint contract makes spatial allocation auditable



Schematic uses public-data proxies; it does not imply statutory planning boundaries.

Figure 1 | Benchmark and Geospatial Kernel execution contract. Public annual land-cover, embedding, night-light, terrain and road layers are aligned to a common 100-m Abu Dhabi grid. Each model receives the same state, action and public proxy exclusions. The Geospatial Kernel exposes a learned proposal, constraint projection and state writeback. Historical, open-loop and planning tracks use one audit contract. The schematic does not imply statutory planning boundaries.

Historical allocation skill is horizon-dependent

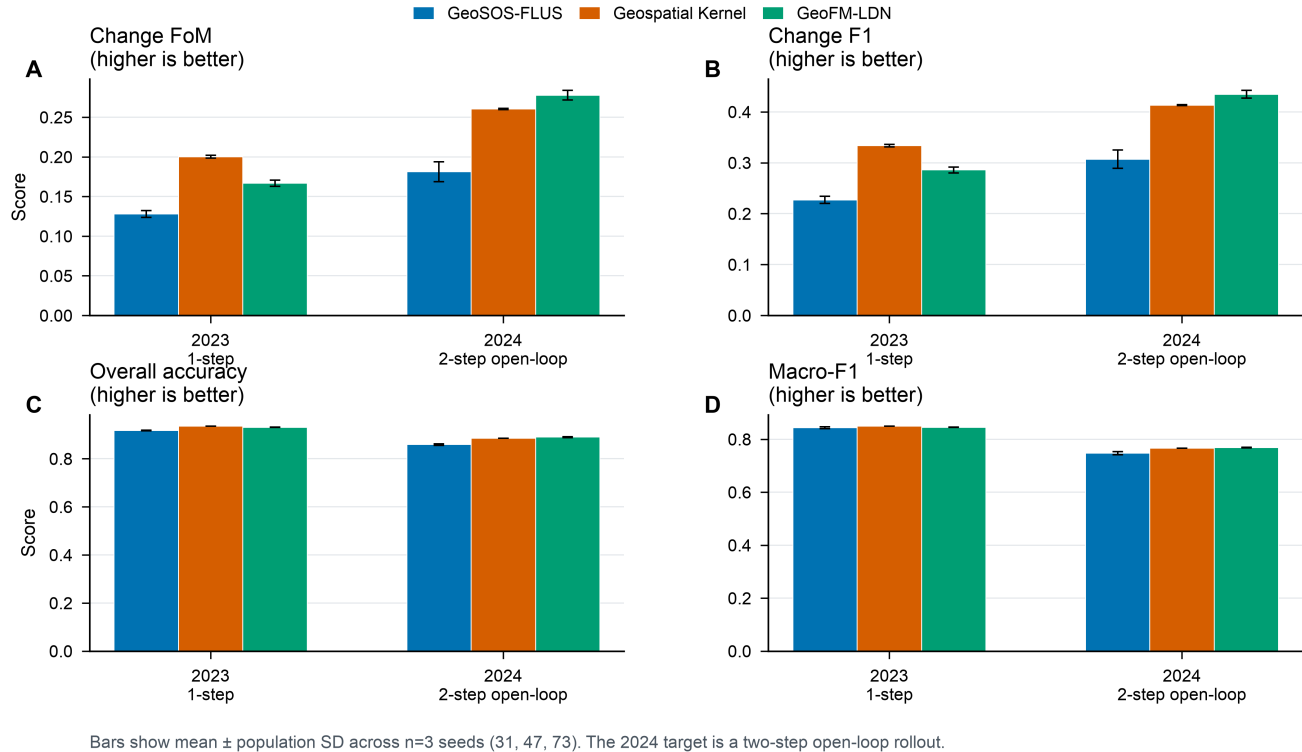


Figure 2 | Historical validation across horizons (legacy, not submission-ready). Strict multi-class change FoM, binary change FoM, change F1, overall accuracy and macro-F1 for the 2023 one-step target and 2024 two-step open-loop target. Persistence and random-feasible zero models are required controls; the current plotted values are legacy artifacts pending strict-metric and bootstrap reruns and must not be used as final evidence.

Geospatial Kernel yields feasible, compact 2031 allocations under three actions

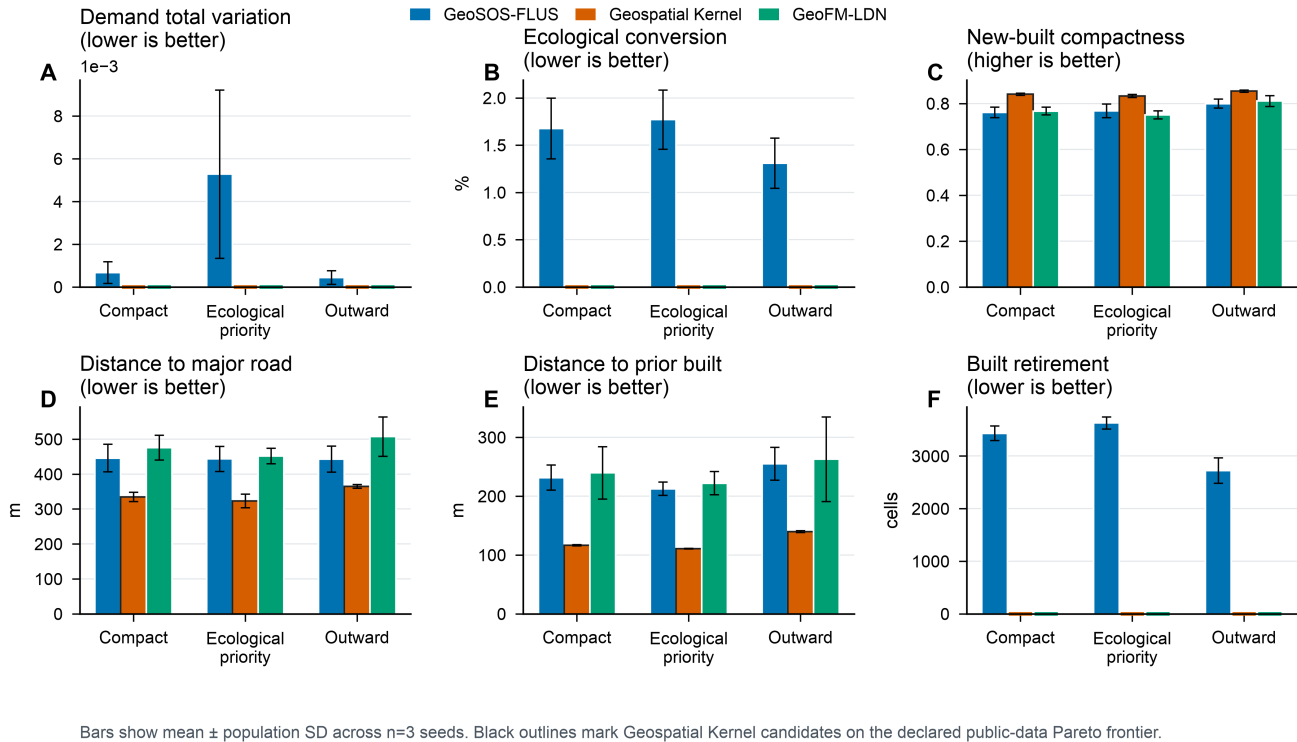
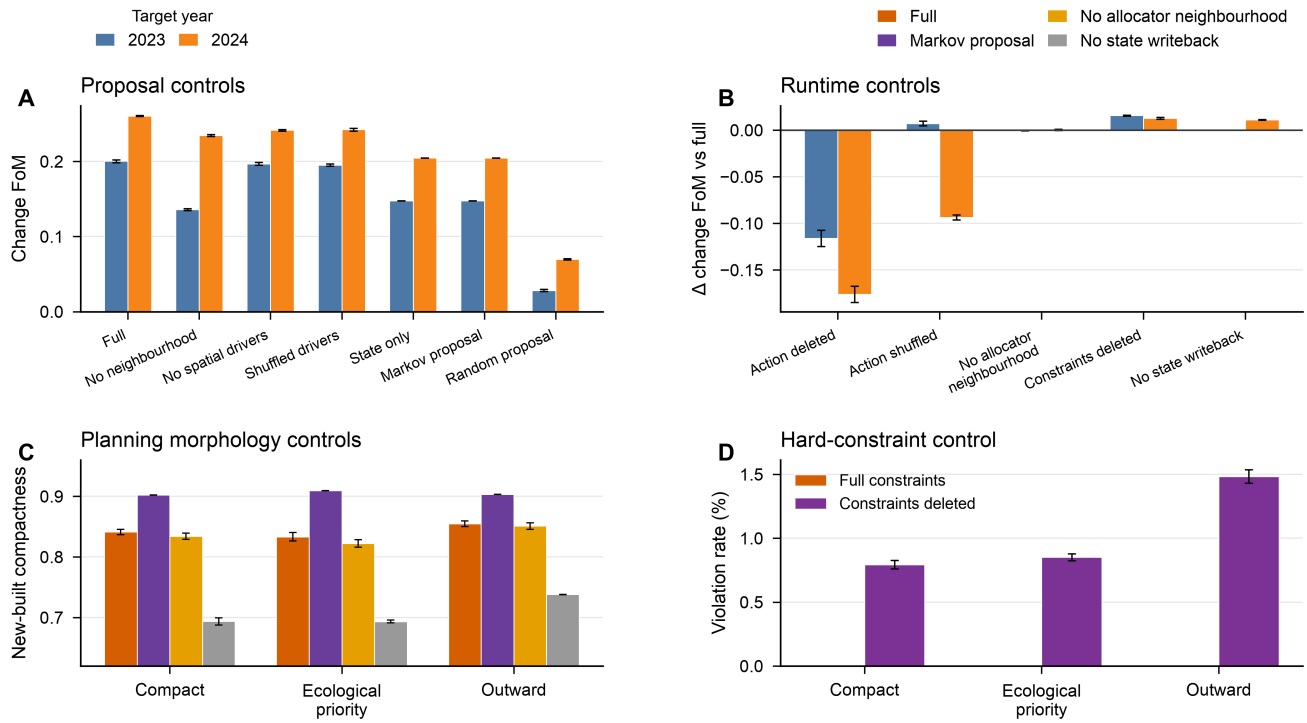


Figure 3 | Conditional planning outcomes at 2031 (legacy, not submission-ready). Descriptive demand and ecological-proxy outcomes are shown alongside independent morphology diagnostics: built components per 1,000 valid cells, 500-m leapfrog rate, distances to major roads and prior built cells, and built retirement. Bars show mean $\pm$ population SD over n=3 seeds. Pareto outlines will be drawn only after the revised four-objective compiler is rerun; the current plotted values are legacy diagnostics.

Mechanism controls separate proposal features from runtime projection



Public-data controls; bars show mean $\pm$ population SD across $n=3$ seeds. These comparisons support, but do not identify, causal mechanisms.

Figure 4 | Mechanism controls (legacy, not submission-ready). Proposal ablations, runtime-control changes relative to the full Kernel, planning compactness controls and hard-constraint violations. These are public-data mechanism controls, not causal policy experiments; the Markov proposal demonstrates that planning morphology is not attributable to the learned proposal alone. All numerical panels require regeneration with the strict evaluator.

Different models produce distinct 2031 transition footprints under identical scenario targets

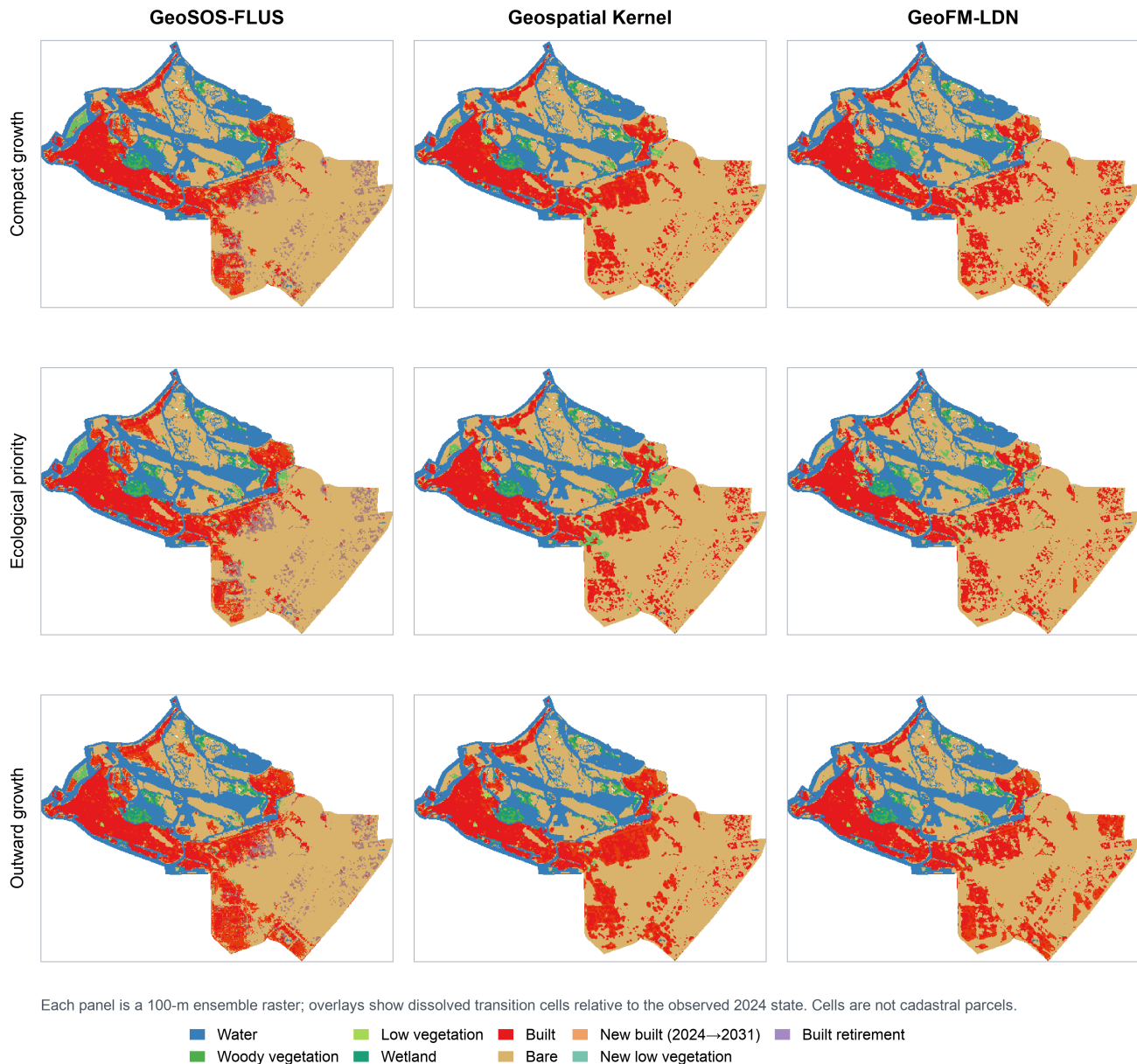


Figure 5 | Spatial allocations and transition footprints at 2031 (legacy, not submission-ready). Rows show the archived moderate-growth (legacy label “Compact”), green-priority-growth (legacy label “Ecological priority”) and high-outward-growth (“Outward”) actions; columns show GeoSOS-FLUS, Geospatial Kernel and GeoFM-LDN. Each panel is a 100-m ensemble raster. Overlays mark new built cells, new low-vegetation cells and built retirement relative to 2024. Transition polygons derived from these cells are not cadastral parcels. The archived artwork predates the planned scale-bar and north-arrow additions and has not been regenerated from the audited ensemble rasters; it must not be used as a final submission figure.

Supplementary information outline

- **Supplementary Note 1:** Complete data crosswalk, quality metrics and public/proxy status.
- **Supplementary Note 2:** Kernel runtime schemas, audit fields and adapter boundary.

- **Supplementary Table S1:** Full 2025–2031 model–scenario objective matrix with per-seed values.
 - **Supplementary Table S2:** Neighbourhood-weight sensitivity at 0, 0.175, 0.35 and 0.7.
 - **Supplementary Table S3:** Raster and vector delivery audit, hashes and layer counts.
 - **Supplementary Figure S1:** Dynamic World confidence and high-confidence change sensitivity.
 - **Supplementary Figure S2:** Historical maps and change-error maps for 2024.
 - **Supplementary Figure S3:** Driver layers and experiment design in English.
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